

Girraween High School

Year 12 Mathematics Extension 2

Task 4

June, 2003

Time 80 minutes

Instructions:

Complete the test on your own paper.

Show all necessary working.

Marks will be deducted for careless or badly arranged work.

Question 1. (43 marks)

a) Find:

(i) $\int \frac{\cos^2 x}{1 - \sin x} dx$

(ii) $\int \frac{x^2 + 2x + 3}{x - 1} dx$ 8

b) Find:

(i) $\int \frac{1}{1 - \cos x} dx$

(ii) $\int \frac{1}{3 + 2x + x^2} dx$ 8

c) Find

(i) $\int_e^{e^2} \ln x dx$

(ii) $\int_0^1 \sqrt{4 - x^2} dx$ 8

d) Given that $\int_0^a f(x) dx = \int_0^a f(a - x) dx$ show that $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$ 5

e) For $f(x) = \frac{20}{(x+1)(x^2+4)}$:

(i) Decompose into partial fractions. 3

(ii) Hence show that $\int_0^2 f(x) dx = \frac{\pi}{2} + \ln\left(\frac{81}{4}\right)$ 4

f) If $I_n = \int_0^t \frac{1}{(1+x^2)^n} dx$ $n = 1, 2, 3, \dots$

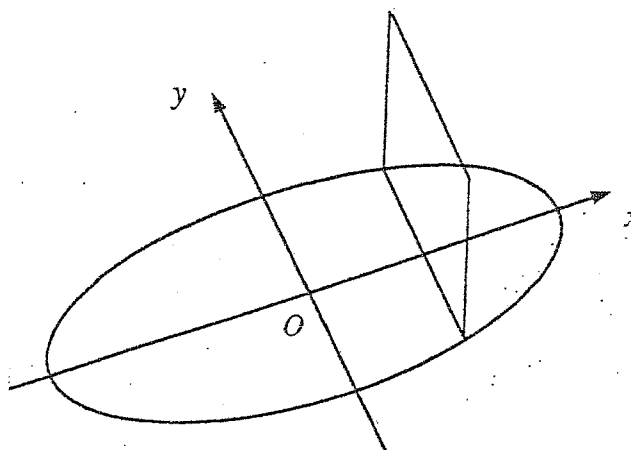
(i) Show $2nI_{n+1} = (2n-1)I_n + \frac{t}{(1+t^2)^n}$ $n = 1, 2, 3, \dots$ 5

(ii) Hence find the value of I_3 , in terms of t . 2

Question 2. (18 marks)

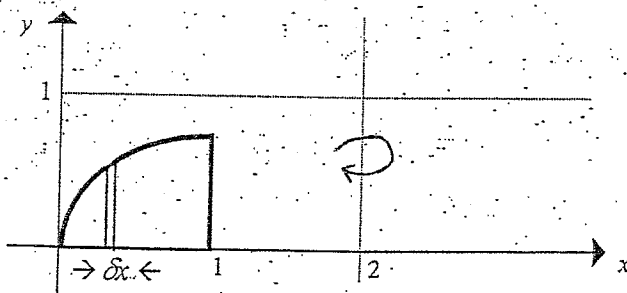
- a) The region bounded by $y = x^2$, the y axis and the line $x = 1$ is rotated about the line $y = -1$. Find the volume of the solid of revolution. 5

- b) The base of a tent is in the shape of an ellipse with the equation $\frac{x^2}{4} + y^2 = 1$. Vertical cross sections taken perpendicular to the major axis are the base of squares.



- (i) Show the volume of the tent is given by $V = \int_{-2}^2 (4 - x^2) dx$. 4
- (ii) Hence find the volume. 2

- c) The region shown bounded by the portion of the curve $y = \frac{x}{x+1}$, the x axis and the line $x = 1$ is rotated about the line $x = 2$.



- (i) Using the method of cylindrical shells, show that the volume δV of a typical shell at a distance x from the origin and with thickness δx is given by 3

$$\delta V = 2\pi(2-x) \cdot \frac{x}{1+x} \cdot \delta x$$

- (ii) Hence find the volume of the solid. 4

Question 1

$$\begin{aligned}
 \text{a) i)} \quad & \int \frac{\cos^2 x}{1 - \sin x} dx \\
 &= \int \frac{(1 - \sin x)(1 + \sin x)}{1 - \sin x} dx \\
 &= \int (1 + \sin x) dx \\
 &= x - \cos x + C
 \end{aligned}$$

$$\text{ii)} \quad \int \frac{x^2 + 2x + 3}{x - 1} dx$$

$$\begin{array}{r}
 x+3 \\
 x-1 \overline{) x^2+2x+3} \\
 \underline{x^2-x} \\
 3x+3 \\
 \underline{3x-3} \\
 6
 \end{array}$$

$$\begin{aligned}
 \therefore &= \int \left(x + 3 + \frac{6}{x-1} \right) dx \\
 &= \frac{x^2}{2} + 3x + 6 \ln(x-1) + C
 \end{aligned}$$

$$\text{b i)} \quad \int \frac{1}{1 - \cos x} dx$$

If $t = \tan \frac{x}{2}$

$$\begin{aligned}
 \frac{dt}{dx} &= \sec^2 \frac{x}{2} \cdot \frac{1}{2} \\
 &= \frac{1 + \tan^2 \frac{x}{2}}{2} \\
 &= \frac{1 + t^2}{2}
 \end{aligned}$$

$$\therefore dx = \frac{2 dt}{1 + t^2}$$

$$\begin{aligned}
 \therefore \int \frac{1}{1 - \cos x} dx &= \int \frac{1}{1 - \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2} \\
 &= \int \frac{1+t^2}{1+t^2 - (1-t^2)} \cdot \frac{2 dt}{1+t^2} \\
 &= \int \frac{2 dt}{2t^2} \\
 &= \int \frac{dt}{t^2} \\
 &= -\frac{1}{t} + C \\
 &= -\cot\left(\frac{t}{2}\right) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad & \int \frac{1}{3 + 2x + x^2} dx \\
 &= \int \frac{1}{2 + (1+x)^2} dx \\
 &= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{1+x}{\sqrt{2}}\right) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{c i)} \quad & \int_e^{e^2} \ln x \cdot \frac{dy}{dx} dx \\
 &= [x \ln x]_e^{e^2} - \int_e^{e^2} x \cdot \frac{1}{x} dx \\
 &= [x \ln x]_e^{e^2} - [x]_e^{e^2} \\
 &= [x \ln x - x]_e^{e^2} \\
 &= (e^2 \ln e^2 - e^2) - (e \ln e - e)
 \end{aligned}$$

$$= e^2 \cdot 2 \ln e - e^2 - e \ln e + e$$

$$\text{Now } \ln e = 1$$

$$\therefore = 2e^2 - e^2 - e + e$$

$$= e^2$$

$$\text{ii)} \quad \int_0^1 \sqrt{4 - x^2} dx$$

$$\begin{aligned}
 \text{let } x &= 2 \sin u \\
 \frac{dx}{du} &= 2 \cos u \\
 dx &= 2 \cos u du
 \end{aligned}$$

$$\begin{aligned}
 x=1 & \quad u = \pi/6 \\
 x=0 & \quad u = 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore & \int_0^{\pi/6} \sqrt{4 - 4 \sin^2 u} \cdot 2 \cos u du \\
 &= \int_0^{\pi/6} 2 \sqrt{1 - \sin^2 u} \cdot 2 \cos u du \\
 &= 2 \int_0^{\pi/6} 2 \cos^2 u du
 \end{aligned}$$

$$\text{Now } 2 \cos^2 u = \cos 2u + 1$$

$$\begin{aligned}
 \therefore & 2 \int_0^{\pi/6} (\cos 2u + 1) du \\
 &= 2 \left[\frac{\sin 2u}{2} + u \right]_0^{\pi/6} \\
 &= 2 \left[\frac{\sin \pi/3}{2} + \frac{\pi}{6} - 0 \right] \\
 &= 2 \left[\frac{\sqrt{3}}{4} + \frac{\pi}{6} \right] \\
 &= \frac{\sqrt{3}}{2} + \frac{\pi}{3}
 \end{aligned}$$

$$\text{d)} \quad \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$$\text{Now } \sin(\pi - x) = \sin x$$

$$\cos(\pi - x) = -\cos x$$

$$\begin{aligned}
 \therefore &= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + (-\cos x)^2} dx \\
 &= \int_0^{\pi} \frac{\pi \sin x - x \sin x}{1 + \cos^2 x} dx \\
 &= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\
 \therefore & 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx
 \end{aligned}$$

$$\text{Now if } u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$\therefore -du = \sin x dx$$

$$x=0 \quad u=1$$

$$x=\pi \quad u=-1$$

$$\begin{aligned} \therefore &= \pi \int_{-1}^1 \frac{-du}{1+u^2} \\ &= \pi \int_{-1}^1 \frac{du}{1+u^2} \\ &= \pi [\tan^{-1}u]_{-1}^1 \\ &= \pi [\tan^{-1}(1) - (\tan^{-1}(-1))] \\ &= \pi [\frac{\pi}{4} - (-\frac{\pi}{4})] \\ &= \pi \cdot \frac{\pi}{2} \\ &= \frac{\pi^2}{2} \end{aligned}$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = \frac{\pi^2}{4}$$

e) $f(x) = \frac{20}{(x+1)(x^2+4)}$

$$\frac{20}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$$

$$\therefore 20 = A(x^2+4) + (Bx+C)(x+1)$$

If $x = -1$

$$20 = 5A + 0$$

$$A = 4$$

If $x = 0$

$$20 = 4A + C$$

$$\therefore C = 4 \text{ as } A = 4$$

If $x = 1$

$$20 = 5A + (8+4) \cdot 2$$

$$20 = 20 + (B+4) \cdot 2$$

$$\therefore B = -4$$

$$\therefore f(x) = \frac{4}{x+1} + \frac{(-4x+4)}{x^2+4}$$

(3)

$$\begin{aligned} \text{ii). } \int_0^2 f(x) dx &= \int_0^2 \frac{4}{x+1} - \frac{(4x-4)}{x^2+4} dx \\ &= \left[4 \ln(x+1) \right]_0^2 - \int_0^2 \frac{4x}{x^2+4} dx + \int_0^2 \frac{4}{x^2+4} dx \\ &= [4 \ln 3 - 0] - 2 \left[\ln(x^2+4) \right]_0^2 + 2 \left[\tan^{-1} \frac{x}{2} \right]_0^2 \\ &= 4 \ln 3 - 2 [\ln 8 - \ln 4] + 2 [\tan^{-1} 1 - \tan^{-1} 0] \end{aligned}$$

$$\begin{aligned} &= 4 \ln 3 - 2 \cdot \ln 2 + 2 \left(\frac{\pi}{4} - 0 \right) \\ &= 4 \ln 3 - 2 \ln 2 + \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} &= \frac{\pi}{2} + \ln 3^4 - \ln 2^2 \\ &= \frac{\pi}{2} + \ln \left(\frac{81}{4} \right) \end{aligned}$$

$$\begin{aligned} \text{f) i) } \int_0^t \frac{1}{(1+x^2)^n} dx \\ I_n = \int_0^t \frac{1}{(1+x^2)^n} dx \\ = \int_0^t \frac{1}{(1+x^2)^n} \cdot \frac{d(x)}{dx} dx \end{aligned}$$

$$= \left[\frac{x \cdot 1}{(1+x^2)^n} \right]_0^t - \int_0^t x \cdot -n(1+x^2)^{-n-1} \cdot 2x dx$$

$$= \frac{t}{(1+t^2)^n} + 2n \int_0^t \frac{x^2}{(1+x^2)^{n+1}} dx$$

$$I_n = \frac{t}{(1+t^2)^n} + 2n \int_0^t \frac{1+x^2-1}{(1+x^2)^{n+1}} dx$$

$$= \frac{t}{(1+t^2)^n} + 2n \int_0^t \frac{1}{(1+x^2)^n} dx - 2n \int_0^t \frac{1}{(1+x^2)^{n+1}} dx$$

$$\therefore I_n = \frac{t}{(1+t^2)^n} + 2n I_n - 2n I_{n+1}$$

$$2n I_{n+1} = 2n I_n - I_n + \frac{t}{(1+t^2)^n}$$

$$2n I_{n+1} = (2n-1) I_n + \frac{t}{(1+t^2)^n}$$

ii) Now $n=2$ gives

$$4 I_3 = 3 I_2 + \frac{t}{(1+t^2)^2}$$

Now $n=1$ gives

$$2 I_2 = I_1 + \frac{t}{1+t^2}$$

$$\begin{aligned} \text{but } I_1 &= \int_0^t \frac{1}{1+x^2} dx \\ &= [\tan^{-1}x]_0^t \\ &= \tan^{-1}t \end{aligned}$$

$$\therefore 2 I_2 = \tan^{-1}t + \frac{t}{1+t^2}$$

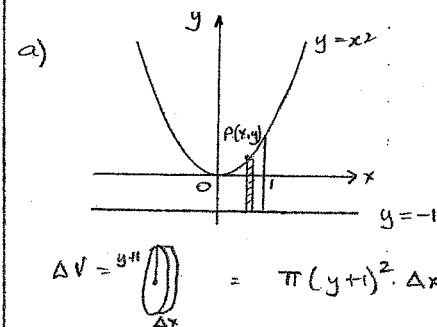
$$I_2 = \frac{1}{2} \left(\tan^{-1}t + \frac{t}{1+t^2} \right)$$

Now

$$4 I_3 = 3 \cdot \frac{1}{2} \left[\tan^{-1}t + \frac{t}{1+t^2} \right] + \frac{t}{(1+t^2)^2}$$

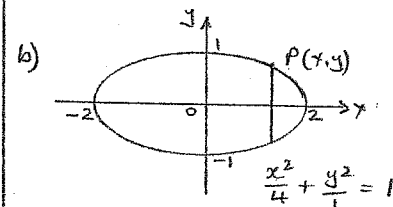
$$\therefore I_3 = \frac{3}{8} \tan^{-1}t + \frac{3t}{8(1+t^2)} + \frac{t}{4(1+t^2)^2}$$

Question 2



$$\begin{aligned} \text{Volume} &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n \pi (y_i+1)^2 \Delta x \\ &= \pi \int_0^1 (y+1)^2 dx \end{aligned}$$

$$\begin{aligned} y+1 &= x^2+1 \\ \therefore &= \pi \int_0^1 (x^2+1)^2 dx \\ &= \pi \int_0^1 (x^4 + 2x^2 + 1) dx \\ &= \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1 \\ &= \pi \left[\frac{1}{5} + \frac{2}{3} + 1 - 0 \right] \\ &= \pi \left[\frac{3+10+15}{15} \right] \\ &= \frac{28\pi}{15} \text{ unit}^3 \end{aligned}$$



$$\Delta V = \pi y^2 \Delta x = 4y^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{-2}^2 4y^2 \Delta x$$

$$= \int_{-2}^2 4y^2 dx$$

$$\text{Now } y^2 = 1 - \frac{x^2}{4}$$

$$\therefore V = \int_{-2}^2 4 \left(1 - \frac{x^2}{4}\right) dx$$

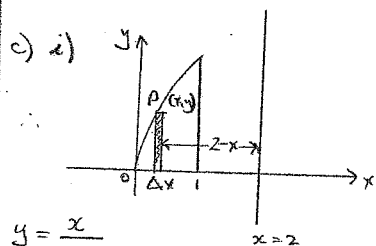
$$V = \int_{-2}^2 ((4 - x^2) dx$$

$$ii) V = \left[4x - \frac{x^3}{3} \right]_{-2}^2$$

$$= \left[\left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) \right]$$

$$= \frac{16}{3} + \frac{16}{3}$$

$$= \frac{32}{3} \text{ or } 10\frac{2}{3} \text{ unit}^3$$



$$\Delta V = y \cdot 2\pi(2-x) \Delta x$$

$$= 2\pi(2-x) \cdot y \cdot \Delta x$$

$$= 2\pi(2-x) \cdot \frac{x}{1+x} \cdot \Delta x$$

$$ii) V = \lim_{\Delta x \rightarrow 0} \sum_0^1 2\pi(2-x) \cdot \frac{x}{1+x} \Delta x$$

$$= 2\pi \int_0^1 \frac{(2-x)x}{1+x} dx$$

$$= 2\pi \int_0^1 \frac{-x^2 + 2x}{x+1} dx$$

$$\begin{array}{r} -x+3 \\ x+1 \overline{) -x^2+2x+0} \\ \underline{-x^2+x} \\ 3x+0 \\ \underline{3x+3} \\ -3 \end{array}$$

$$\therefore = 2\pi \int_0^1 \left(-x+3 - \frac{3}{x+1}\right) dx$$

$$= 2\pi \left[-\frac{x^2}{2} + 3x - 3\ln(x+1) \right]_0^1$$

$$= 2\pi \left[\left(-\frac{1}{2} + 3 - 3\ln 2\right) - (0 + 0 - 3\ln 1) \right]$$

$$= 2\pi \left[-\frac{1}{2} + 3 - 3\ln 2 \right]$$

$$= \pi \left[-1 + 6 - 6\ln 2 \right]$$

$$= \pi \left[5 - 6\ln 2 \right]$$

$$\text{or } 5\pi - 6\pi \ln 2$$